DRISTII PATEL

BS (Research) Physics · Indian Institute of Science (IISc), Bengaluru

Neutrino physics · Particle detectors · Computational astroparticle physics

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SUMMARY

Third-year BS (Research) Physics student at IISc working on neutrino physics and computational astroparticle physics under Dr. Biswaranjan Behera. My work covers supernova and solar neutrino detection, sterile-neutrino phenomenology and detector sensitivity modelling, and a manuscript on sterile neutrinos in solar ^{8}B neutrinos is in preparation. I have reproduced published calculations without the original code, built numerical analysis pipelines, and contribute to the open-source SNEWPY package through the SNEWS2 collaboration. Looking for research internships now, and PhD programmes in neutrino and particle physics from 2028.

EDUCATION

Bachelor of Science (Research), Physics 2024 – 2028
Indian Institute of Science (IISc), Bengaluru, India (NIRF 2025: #1 University · QS World 2027: #221)

- **CGPA 8.3/10**; fourth-semester TGPA 9.0/10
- **Admission:** JEE Advanced 2024, **AIR 1925** (of ∼180,000 candidates); JEE Main 2024, AIR 3898 (99.76 percentile)
- **Coursework:** Intermediate Classical Mechanics; Intermediate Electromagnetism & Quantum Physics of Radiation; Intermediate Thermal Physics & Materials; Introduction to Quantum Measurements

Higher Secondary (GSEB), Crystal School 86.78% (Physics 91, Mathematics 96) 2022 – 2024

RESEARCH EXPERIENCE

Undergraduate Researcher | Supernova & Solar Neutrino Detection 2025 – Present
Centre for High Energy Physics, IISc | Supervisor: Dr. Biswaranjan Behera (SERB Ramanujan Fellow)

Computational modelling of supernova and solar neutrino signals in deuterated liquid scintillator (DLS) detectors: event-rate calculations, detector-response modelling and sensitivity analysis.

Reproduction of DLS supernova-neutrino event rates (Chauhan, Dasgupta & Datar, JCAP 2021) [code]

- Rebuilt the event-rate pipeline from the published description, without the original code (fluence, digitised NSGK deuteron cross sections, Birks'-law quenching, CC/NC/ES spectra); **NC rates within 1–3%** of the published tables
- Investigated an initial 17–20% CC discrepancy: ruled out coding errors with an independent NumPy re-implementation, traced it to coarse sampling in the printed cross-section tables, and resolved it to **<4%** using the original NSGK tables
- Extended beyond the paper: recoil-neutron spectra from EXFOR data, secondary breakups, deuteron secondary scintillation

Neutrino mass-ordering sensitivity across eight detector configurations [code]

- Developed a Poisson-significance framework with adiabatic MSW conversion to compare mass-ordering sensitivity across eight detectors (H_2O, D_2O, WbLS, Gd/Te-doped, D_2O+LAB) in IBD, CC, NC and ES channels; D_2O rates agree with published values to ∼1%
- Used time-dependent SNEWPY fluxes (Nakazato, Bollig, Tamborra) with adaptive time-binning to recover accretion-phase sensitivity lost in whole-burst integration; **identified the choice of supernova simulation as the dominant source of sensitivity variation**

Solar ^{8}B neutrino survival probability: sterile-neutrino sensitivity [code] [summary] *Manuscript in preparation*

- Computed $P_{ee}(E)$ for three-flavour LMA-MSW and 3+1 sterile-neutrino models by numerically evolving the Hamiltonian through realistic solar density profiles; identified the energy window of maximum model separation and the exposure needed to discriminate them
- Day–night asymmetry with the PREM Earth density profile; CC spectra and energy-dependent asymmetry for four DLS configurations in a 1 kton detector

OPEN-SOURCE SOFTWARE

SNEWPY (SNEWS2 Collaboration) : Diffuse Supernova Neutrino Background (DSNB) Module 2026 – Present
*Python, NumPy, SciPy · github.com/SNEWS2/snewpy · **PR #453 under review for SNEWPY 2.0***

- Implemented redshift-dependent DSNB flux calculations from cosmological core-collapse rates and IMF-weighted progenitor spectra, with per-progenitor MSW flavour transformations supporting both mass orderings; validated against published DSNB calculations.
- Integrated with SNEWPY's supernova-model, Flux/Fluence, flavour-transformation and RateCalculator APIs; maintained a **28-test suite** through iterative refactoring and maintainer feedback.
- Contributing to SNEWPY documentation and reviewing existing tutorials and notebooks, identifying inconsistencies and usability issues from a new-contributor perspective.

SELECTED RESEARCH PROJECTS

DSNB Detection Sensitivity: Reproduction of Li, Vagins & Wurm (2022) 2025 – 2026
Python, NumPy, SciPy, Matplotlib · gitlab.com/dristiipatel/dsnb-reproduction

- Reproduced Table III (SK-Gd and JUNO) **to rounding precision** from first principles (arXiv:2201.12920): DSNB flux over the evolving core-collapse rate, Strumia–Vissani IBD cross section, detector and background models; vectorised the 2D (E, z) integration for a $\sim$50$\times$ speed-up
- Produced five figures beyond the paper, including a 2D significance map in $(\langle E_\nu \rangle, f_{\mathrm{BH}})$ space; documented the remaining $\sim$15% signal-rate discrepancy with the published calculation rather than correcting it empirically

CID Method Simulation: Directional Solar-Neutrino Extraction (Borexino) 2026
Python, NumPy, SciPy, Matplotlib · gitlab.com/dristiipatel/cid-solar-neutrino-sim

- Simulated Borexino's Correlated and Integrated Directionality technique (Basilico et al., arXiv:2307.14636) with a synthetic PMT-sphere detector (scintillation and directional Cherenkov hits); joint χ^2 fit across Nth-hit $\cos\alpha$ slices with a Bayesian posterior on signal counts
- Calibration check: 68% credible intervals covered the true value in 73% of 30 trials; simplifications relative to the published analysis documented

GW150914 via Matched Filtering 2026
Python, PyCBC, GWOSC · gitlab.com/dristiipatel/gw-matched-filter

- Built the pipeline (GWOSC data, PSD estimation, whitening, IMRPhenomD templates); recovered GW150914 independently in H1 and L1 with peak SNR **17.55** and **13.10**, within $\sim$30 ms of the published merger time

SCHOOLS, WORKSHOPS & TRAINING

PHLEX: A Next-Generation Framework for DUNE, and AI/ML Libraries in HEP [event page] 26–30 October 2026
Rice University Global · Centre for High Energy Physics, IISc

- Attended a workshop on the PHLEX (Parallel, Hierarchical, and Layered Execution) framework for DUNE data-processing algorithms, covering scientific software development and AI/ML libraries for high-energy physics.

Summer School for Women in Physics (SSWP), ICTS Bengaluru June 2025
Intensive research programme

- Experimental methodology and uncertainty analysis; laser-diffraction and Brownian-motion experiments with extracted diffusion coefficients

High Energy Physics & Particle Detectors March 2026
Directed reading project under Dr. Behera, IISc

- Radiation detection, charged-particle energy loss and Cherenkov radiation (Knoll; Garutti's HEP detector lecture notes)

SCIENCE COMMUNICATION

IISc Open Day 2026 : Live Muon Detection Demonstration March 2026
Indian Institute of Science, Bengaluru · public audience

- Set up and operated a live muon-detection experiment, explaining Cherenkov radiation and charged-particle detection to visitors without a physics background

HONORS & AWARDS

Deep Asher Fellowship 2024–25 (IISc Class of 1998 ME(Int)) : one B.Sc student per year, awarded on entrance-exam rank and first-semester TGPA 2024

NSEA 2024 (IAPT) : state-level selection for the Indian National Astronomy Olympiad (INAO) 2024

IAAC 2025 : Round 2 qualifier; Round 1 submission featured on the official website for clarity of reasoning 2025

LABORATORY EXPERIENCE

30+ undergraduate physics experiments at IISc, involving experimental instrumentation, data acquisition, uncertainty analysis and Python-based data processing. Representative experiments include:

- **Optics:** Mach–Zehnder interferometry, Fabry–Perot resonator, Nd:YAG laser, Cornu's method.
- **Modern physics:** Compton scattering, electron diffraction, atomic spectra, Zeeman effect.
- **Electronics and materials:** Hall effect, Faraday effect, B–H hysteresis, Seebeck and Peltier effects, semiconductor I–V.

SKILLS

Scientific computing: Python, NumPy, SciPy, Matplotlib · **Research software:** SNEWPY, PyCBC, SNOwGLoBES · **Tools:** Git, GitHub, GitLab, LaTeX

Methods: Numerical modelling, neutrino oscillation calculations, event-rate simulations, statistical inference, sensitivity analysis, uncertainty quantification, reproduction and validation of published results

Languages: English (professional), Hindi (native), Gujarati (native)

ACTIVITIES & INTERESTS

Ensemble (IISc Physics Club); Samasya (IISc Maths Club); Pravega 2025 event coordination; Sangam performances (keyboard, vocals). *Interests:* keyboard, singing, sketching, journaling.

REFERENCE

Dr. Biswaranjan Behera : research supervisor; SERB Ramanujan Fellow, Centre for High Energy Physics, IISc